

Exploring How Neurofeedback Data Visualization Affects User Immersiveness and Experience in Virtual Reality

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ABSTRACT

Neurofeedback is a form of biofeedback in which a sensory representation of brain activity is fed back to the user in real time to facilitate an understanding of brain function and improve focus and/or attention. Researchers are increasingly using task-related EEG values in training systems for cognitive adaptive development. However, there is a notable lack of research on the means by which EEG data should be visualized to enhance the immersive quality of the experience. In the current study, we plan to qualitatively explore the impact of neurofeedback visualization schemes on the immersiveness of the VR experience. The results of this in-lab study are expected to provide important insights into the design of visualization schemes for emerging VR applications.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Neurofeedback; Virtual reality; Visualization; Focus score.

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1 INTRODUCTION

Neurofeedback refers to the process of feeding a sensory representation of brain activity obtained using an electroencephalograph (EEG) back to the user in real time to facilitate an understanding of brain function and improve focus and/or attention [10]. There is a growing body of evidence suggesting that head-mounted virtual reality (VR) systems are more effective than conventional 2D representations in engaging attentional resources [8]. Despite extensive work on the use of task-related alpha activity in cognitive adaptive training systems [2], there has been a notable lack of research on the means by which EEG data can be visualized without compromising the immersive quality of the experience. An effective strategy for the visualization of EEG data (or inferred data) could facilitate the design of apps that rely on visualizations to refine focus and/or attention.

Neurofeedback is one of many technological innovations that have been used as therapeutic tools for the treatment of conditions, such as attention deficit hyperactivity disorder [4]. VR has proven highly valuable in neurofeedback therapy [9] by providing a highly immersive imaginary world featuring situations that would be difficult to reproduce in the real world or using devices with 2D screens. The development of wearable EEG devices [3] has moved neurofeedback technology out of the lab and into daily life. However, when visualizing these EEG readings in VR worlds, existing research studies have focused on other types of biofeedback data (e.g., HRV or breathing data) [5, 6]. Therefore, we hope to explore how to visualize these EEG data in VR world without sacrificing the immersiveness of the VR world while increasing the VR user experience.

In the current study, we will conduct a within-subject study with the aim of qualitatively exploring how these visualization schemes affect the immersiveness of the VR user experience. Volunteers will provide qualitative opinions of how the schematic visual representations affected their focus. The results of this in-lab study are expected to provide important insights into the design of visualization schemes for emerging VR applications.

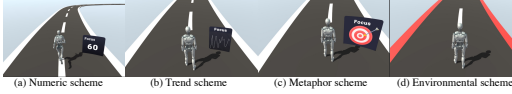


Figure 1: Screenshots of VR App for training focus-related cognitive skills using various visual representations.

2 EEG DATA VISUALIZATION STUDY IN THE VR WORLD

2.1 Device

We will be using an Emotiv EPOC X sensor [3] for the collection of EEG brain signals in real time. Data transmission and analysis aimed at deriving performance metrics of mental load will be handled using EmotivPRO software [3]. Focus scores (ranging from 0 to 100) will be extrapolated from TAR values [7], an important indicator of cognitive load.

2.2 VR App in Training for Improved Focus

The objective of the app (Figure 1) is to enhance the peak performance of the participant by training their ability to focus. While using the app, the participant will focus on moving the legs of their avatar (in the center of the screen) faster without actually performing the movement by themselves (i.e., using only motor imagery). The ability of the participant to making the movement of the legs faster is expected to give an accurate indication as to the level of focus. By displaying the corresponding focus score in the form of a visualization, it should be possible for the participant to gauge their ability to focus on the assigned task and develop this ability in real time.

2.3 Procedure

This within-subject study of the four visualization schemes described earlier will be implemented in three stages: (1) pre-study, (2) VR simulation, and (3) post-study.

2.3.1 Pre-study Stage. The researchers will first explain to the participants the study goals of the study and the procedures used in the in-lab study. Researchers will also assist the participants to put on the VR headset and EEG brain sensing device to ensure that data is reliably collected.

2.3.2 VR Experiencing Stage. This stage will be implemented in the following three phases.

Familiarization phase (5 minutes). In this phase, the objective is to help participants transmission smoothly from the physical to a virtual world. Upon entering the VR world, participants first find themselves in a virtual copy of the meeting room they left in the real world. The participants will then watch a VR tutorial on how to interact with the system using the controllers. By clicking on a button, the participant will begin the visualization phase, in which the four visualization schemes appear in random order.

Visualization phase (7 minutes for each scheme). At the beginning of each session, the participants will be given instructions pertaining to their role in the simulation (i.e., making the avatar run as fast as possible). Note that the first trial of each visualization scheme will be disregarded. Once participants complete an

actual trial of a visualization scheme, a question-answer sequences (Q&A sequences) will immediately be conducted to learn their opinions about the immersiveness and user experience when using the scheme they just completed. Each participant will be expected to complete all four visualization schemes.

Visualization scheme (5 minutes). Our review of the literature [5, 6] revealed four approaches to the visualization of focus scores in the context of VR. The *numeric scheme* (Figure 1(a)) presents real-time values of focus scores to participants as they control the avatar. The *trend scheme* (Figure 1(b)) plots the focus score as a function of time in a line graph, showing the recent history of the scores. The *metaphor scheme* (Figure 1(c)) provides a real-time representation of the focus score as an arrow approaching a bullseye, where the center indicates perfect concentration. The *environmental scheme* (Figure 1(d)) uses a color gradient (0-100%) to represent the focus score, where the degree of saturation corresponds to the degree of concentration. The color appears along the sides of the road.

Question-answer sequences (2 minutes). During the Q&A sequences, the participants will answer questions related to the effect of the visualization on the immersiveness of the experience. Answers will be presented using a visual analogue scale (VAS) or seven-level Likert scale.

Exit phase (one minute). Once again, we want the participants to transition smoothly from the virtual world to the physical world. This will be achieved by reversing the familiarization phase, wherein the last VR image presented to the participant will be a virtual model of the meeting room in which they are currently sitting.

2.3.3 Post-study Stage. After the simulations, a follow-up semi-structured interview will be conducted to obtain information related to the pros and cons of each visualization scheme and the participant's personal preferences. It is also meant to determine the frequency with which the participants checked the focus scores and how they interpreted the visualized data. We will try to determine how the visualized data affected the immersiveness of the experience, and enquire about the lessons they learned while seeking to improve their focus.

2.4 Planned Measures & Data analysis

Measures of immersiveness will be derived from the self-reported recollections (Q&A) presented immediately after each task. Responses during the semi-structured interviews will be recorded and transcribed, before undergoing iterative coding independently by three experienced researchers with the aim of identifying salient themes [1].

3 FUTURE WORK & CONCLUSION

We will recruit participants to collect qualitative opinions towards these visualization schemes. The results of this in-lab study are expected to provide important insights into the design of visualization schemes for emerging VR applications.

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